Yuqi Liang

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Education

University of Electronic Science and Technology of China (UESTC) 2023-Now **Chengdu, China**

- B.S. in Automation, GPA *3.75/4.0*
- Awards: Campus-level Best Cadet, Hong Kong Leadership Training Camp (2024)
  First Prize, Electronic Design Competition, School of Automation Engineering (2023)
  First Prize, CSP-J Group, NOIP (2019)
- Leadership and Extracurricular Activities: Director of External Relations, Commissary in Charge of Studies of Outstanding Class, Member of Class 1 of the "Li ren" Program at UESTC

Academic experience

Auto-Tuning for Data-driven Model Predictive Control

Research Assistant (September 2024 - Present)

- Reproduced the paper and explored alternative heuristic optimization algorithms to replace the single Bayesian optimization algorithm.
- Proposed the Balanced Truncation Model Reduction (BTMR) method in order to better reflect the energy distribution in singular matrix of the system in system identification.
- Utilized Balanced Proper Orthogonal Decomposition (BPOD) and Eigensystem Realization Algorithm (ERA) to resolve matrix adjustment issues highlighted in the research.

Based Genetic Algorithm PID Parameters Regulator

Research Assistant (June 2024 – August 2024)

- Compared heuristic and non-heuristic optimization algorithms for PID parameter tuning, ultimately implementing the Genetic Algorithm (GA).
- Collected offline data and optimized the traditional PID algorithm, leading to a victory in the Intelligent Tracking Car Competition.
- Utilized the inverted pendulum simulation model to compare its performance with other modern control methods.

Automated Tracking Car- Machine Vision-Based Trajectory Recognition

Research Assistant (April 2024 – June 2024)

- Applied computer vision to automatically correct misjudgments in the path identification of the tracking car caused by traditional electromagnetic sensors, and implemented the simulation.
- Identified an issue and proposed a preliminary solution: Due to cost constraints, the single-chip computer's performance cannot support high-performance online computing.
- Employed hyperparameter optimization to determine the car's path through comparison, allowing for further assessment of the need to adjust and tune the PWM duty cycle.

Skills

- Programming Languages：Python, PyTorch, C/C++, MATLAB, LaTeX
- Tools：Microsoft Office, Origin Lab, Overleaf, Microsoft Visio